INFOSYS SYSTEM ENGINEER 1:30Pm Slot

SECTION 1 — REASONING ABILITY TEST

15 Questions | 15 Marks | 25 Minutes

1. In a family, Karan is the brother of Neha. Neha is the daughter of Raj. How is Raj related to Karan?

A) Uncle
B) Father
C) Brother
D) Cousin
Answer: B) Father

2. Five people P, Q, R, S and T sit in a row facing north. Q sits immediately to the right of P. R sits immediately to the left of S. T sits at the extreme right. Who sits in the middle?

A) P
B) Q
C) R
D) S
Answer: B) Q

3. Statements: All programmers are logical thinkers. Some logical thinkers are writers. Conclusions: I. Some programmers are writers. II. Some writers are logical thinkers. Which conclusion(s) logically follow?

A) Only I
B) Only II
C) Both I and II
D) Neither I nor II
Answer: B) Only II

4. A standard die has opposite faces adding up to 7. If 2 is on the top face, which number is on the bottom face?

A) 3
B) 4
C) 5
D) 6
Answer: C) 5

5. Six students A, B, C, D, E and F stand in a line. A is second from the left. B is immediately to the right of A. C is at the extreme right. D is immediately left of E. Who can occupy the third position?

A) B
B) D
C) E
D) F
Answer: A) B

6. A person walks 9 km east, turns left and walks 5 km, then turns left and walks 9 km. Where is the person from the starting point?

A) 5 km north
B) 5 km south
C) 9 km east
D) 14 km north
Answer: A) 5 km north

7. Find the next letter: B, F, J, N, R, __.

A) T
B) U
C) V
D) W
Answer: C) V

8. Choose the odd one out.

A) Mercury
B) Venus
C) Earth
D) Moon
Answer: D) Moon

9. If 1 January 2024 was a Monday, what day was 1 January 2025?

A) Monday
B) Tuesday
C) Wednesday
D) Thursday
Answer: C) Wednesday

10. Find the next number: 4, 9, 16, 25, 36, __.

A) 42
B) 45
C) 49
D) 52
Answer: C) 49

11. A father is three times as old as his son. Ten years ago, he was five times as old as his son. What is the son's present age?

A) 15
B) 20
C) 25
D) 30
Answer: B) 20

12. In a group of 12 people, every person shakes hands exactly once with every other person. How many handshakes occur?

A) 54
B) 60
C) 66
D) 72
Answer: C) 66

13. If yesterday was Sunday, what day will it be the day after tomorrow?

A) Tuesday
B) Wednesday
C) Thursday
D) Friday
Answer: B) Wednesday

14. Find the next number: 5, 11, 19, 29, 41, __.

A) 53

B) 54
C) 55
D) 57
Answer: C) 55

15. Complete the analogy: 6 : 36 :: 9 : ?

A) 54
B) 72
C) 81
D) 90
Answer: C) 81

SECTION 2 — TECHNICAL ABILITY TEST (QUANTITATIVE APTITUDE)

10 Questions | 10 Marks | 35 Minutes

1. Find the 10th term of the arithmetic sequence 7, 12, 17, 22, ...

A) 47
B) 50
C) 52
D) 55
Answer: C) 52

2. A price is increased by 25% and then decreased by 10%. What is the net percentage change?

A) 10% increase
B) 12.5% increase
C) 15% increase
D) 15% decrease
Answer: B) 12.5% increase

3. The average of six numbers is 28. Five numbers are 17, 24, 29, 31 and 35. Find the sixth number.

A) 30
B) 32
C) 34
D) 36
Answer: C) 34

4. A and B invest in the ratio 4:7. If the total profit is ₹8800, what is B's share?

A) ₹3200
B) ₹4800
C) ₹5600
D) ₹6400
Answer: C) ₹5600

5. An article costing ₹1250 is sold for ₹1500. What is the profit percentage?

A) 15%
B) 18%
C) 20%
D) 25%
Answer: C) 20%

6. A train travels 270 km at 90 km/h. How much time does it take?

A) 2 hours
B) 2.5 hours

C) 3 hours
D) 3.5 hours
Answer: C) 3 hours

7. Pipe A fills a tank in 15 hours and Pipe B fills it in 10 hours. How long will they take together?

A) 5 hours
B) 6 hours
C) 7 hours
D) 8 hours
Answer: B) 6 hours

8. In how many ways can 6 distinct students be arranged in a row?

A) 120
B) 240
C) 360
D) 720
Answer: D) 720

9. From 8 people, in how many ways can a committee of 3 be selected?

A) 24
B) 48
C) 56
D) 64
Answer: C) 56

10. If CAT + DOG is a word-number puzzle where each distinct letter represents a distinct digit, which of the following could be the value of CAT?

A) 124
B) 135
C) 246
D) 357
Answer: B) 135

SECTION 3 — VERBAL ABILITY TEST

20 Questions | 20 Marks | 20 Minutes

1. Choose the word closest in meaning to “meticulous”.

A) Careless
B) Precise
C) Impulsive
D) Casual
Answer: B) Precise

2. Choose the word closest in meaning to “obsolete”.

A) Modern
B) Useful
C) Outdated
D) Flexible
Answer: C) Outdated

3. Choose the word opposite in meaning to “scarce”.

A) Rare
B) Limited

C) Abundant
D) Insufficient
Answer: C) Abundant

4. Choose the word closest in meaning to “pragmatic”.

A) Practical
B) Emotional
C) Theoretical
D) Careless
Answer: A) Practical

5. Choose the word closest in meaning to “ambiguous”.

A) Certain
B) Unclear
C) Honest
D) Simple
Answer: B) Unclear

6. What does the idiom “spill the beans” mean?

A) Waste food
B) Reveal a secret
C) Make a mistake
D) Start a fight
Answer: B) Reveal a secret

7. What does “under the weather” mean?

A) Feeling unwell
B) Feeling excited
C) Working outdoors
D) Being late
Answer: A) Feeling unwell

8. What does “a blessing in disguise” mean?

A) A hidden problem
B) Something that seems bad but turns out beneficial
C) A public celebration
D) An obvious advantage
Answer: B) Something that seems bad but turns out beneficial

9. What does “call it a day” mean?

A) Begin work
B) Stop working for the day
C) Change a schedule
D) Make a phone call
Answer: B) Stop working for the day

10. What does “on the same page” mean?

A) Reading together
B) Having the same understanding
C) Disagreeing strongly
D) Working separately
Answer: B) Having the same understanding

11. Choose the correct option: They ____ dinner when the guests arrived.

A) have
B) were having
C) will have
D) had have
Answer: B) were having

12. Choose the correct option: If she ____ harder, she would have passed the exam.

A) studies
B) studied
C) had studied
D) has studied
Answer: C) had studied

13. By next month, he ____ at the company for two years.

A) works
B) worked
C) will have worked
D) has worked
Answer: C) will have worked

14. She ____ this book three times already.

A) reads
B) read
C) has read
D) is reading
Answer: C) has read

15. We cancelled the match ____ it was raining heavily.

A) although
B) because
C) unless
D) so that
Answer: B) because

16. Identify the part containing the error: “Each of the employees have submitted the report.”

A) Each of the employees
B) have submitted
C) the report
D) No error
Answer: B) have submitted

17. Identify the part containing the error: “She is senior than me in the organization.”

A) She is
B) senior than
C) me
D) in the organization
Answer: B) senior than

18. Identify the part containing the error: “I have completed the work yesterday.”

A) I have
B) completed
C) the work
D) yesterday
Answer: A) I have

19. Identify the part containing the error: “Between you and I, this plan needs revision.”

A) Between you and I
B) this plan
C) needs
D) revision
Answer: A) Between you and I

20. Identify the part containing the error: “He is one of the best player in the team.”

A) He is
B) one of the best
C) player
D) in the team
Answer: C) player

SECTION 4 — PSEUDO CODE TEST

5 Questions | *10 Marks* | *10 Minutes*

1. Consider the following pseudocode:

```
A = [2, 5, 7, 9]
FOR i = 0 TO 2
  A[i] = A[i] + A[i+1]
END FOR
PRINT A
```

What is the output?

A) [7, 12, 16, 9]
B) [7, 14, 16, 9]
C) [2, 7, 12, 16]
D) [7, 12, 16, 18]
Answer: A) [7, 12, 16, 9]

2. Consider the following pseudocode:

```
result = 1
FOR i = 1 TO 5
  IF i MOD 2 != 0
    result = result * i
  END IF
END FOR
PRINT result
```

What is the output?

A) 9
B) 15
C) 30
D) 45
Answer: C) 15

3. Consider the following pseudocode:

```
FUNCTION G(n)
  IF n <= 1
    RETURN n
  ELSE
    RETURN G(n-1) + G(n-2)
  END IF
```

END FUNCTION
PRINT G(6)
What is the output?

A) 5
B) 8
C) 13
D) 21
Answer: B) 8

4. Consider the following pseudocode:

```
FOR i = 1 TO N
  FOR j = 1 TO i
    PRINT i + j
  END FOR
END FOR
```

What is the time complexity in terms of N?

A) O(N)
B) O(log N)
C) O(N^2)
D) O(2N)
Answer: C) O(N^2)

5. Consider the following pseudocode:

```
count = 0
FOR EACH x IN [4, 9, 12, 15, 18, 21]
  IF x MOD 3 == 0
    count = count + 1
  END IF
END FOR
PRINT count
```

What is printed?

A) 3
B) 4
C) 5
D) 6
Answer: C) 4

SECTION 5 — NUMERICAL PUZZLE TEST

4 Questions | 10 Marks | 10 Minutes

1. Maze/Grid Puzzle:
A robot starts at S and must reach E. It can move only UP, DOWN, LEFT or RIGHT and cannot enter cells marked X.
Grid:

```
S . . X .
X . . X .
. . X . .
. X . . .
. . . X E
```

What is the minimum number of moves required to reach E from S?

A) 8
B) 9

C) 10
D) 11
Answer: B) 9

2. Matrix Puzzle:
Find the missing number (?) in the matrix.
3 7 11
5 10 15
7 13 ?
Choose the number that logically completes the pattern.

A) 17
B) 18
C) 19
D) 20
Answer: C) 19

3. Number-Path Puzzle:
Starting from S(2), move horizontally or vertically to an adjacent cell. You may enter a numbered cell only if its number is greater than the number in the cell currently occupied.
Grid:
S(2) 4 7 9
3 5 8 10
1 6 11 12
0 7 13 E(14)
Which gives a valid path from S to E?

A) 2→4→7→8→11→12→14
B) 2→3→5→6→11→13→14
C) 2→4→5→8→6→13→14
D) 2→7→8→10→12→14
Answer: A) 2→4→7→8→11→12→14

4. Missing-Cell Grid Puzzle:
Each row follows the same numerical rule.
4 8 12
6 12 18
9 18 ?
What number should replace ?

A) 24
B) 25
C) 27
D) 30
Answer: C) 27

SECTION 6 — ENGLISH GRAMMAR TEST

5 Questions | 10 Marks | 10 Minutes

1. Neither of the answers ____ correct.

A) are
B) were
C) is
D) have
Answer: C) is

2. The manager, along with his assistants, ____ attending the conference.

A) are
B) were
C) is
D) have
Answer: C) is

3. She made me ____ the report again.

A) to check
B) checking
C) check
D) checked
Answer: C) check

4. The letters ____ by the secretary were sent yesterday.

A) typed
B) was typed
C) typing
D) is typed
Answer: A) typed

5. Every student in the class ____ required to carry an ID card.

A) are
B) have
C) were
D) is
Answer: D) is

SECTION 7 — ENGLISH WRITTEN TEST

1 Question | Evaluated | 10 Minutes

1. Write an essay of approximately 150–200 words on the following topic:
“Technology can improve productivity, but human skills remain essential in the modern workplace.”